



AMRITA SCHOOL OF ENGINEERING, BENGALURU

**DEPARTMENT OF ELECTRONICS & COMMUNICATION
ENGINEERING**

B. TECH. IV SEMESTER (EAC) AY 2025-2026

**ASSIGNMENT FOR 23ECE211 MICROCONTROLLERS AND
INTERFACING**

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**SYSTEM DESIGN FOR 23ECE284 MICROCONTROLLERS AND
INTERFACING LABORATORY**

REPORT

JANUARY TO APRIL 2026

Smart Health Monitoring System using LPC2148

Submitted by Padmaja

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BATCH NUMBER: 10

FACULTY EXAMINER SIGNATURE WITH EVALUATION DATE:

Smart Health Monitoring System using LPC2148

CIRCUIT DIAGRAM:

CONNECTIONS:

LCD (16x2, 4-bit mode) → Port 0

- RS → P0.16
- RW → P0.17
- EN → P0.18
- D4 → P0.19
- D5 → P0.20
- D6 → P0.21
- D7 → P0.22

LEDs → Port 1

- P1.16 to P1.23 → 8 LEDs (Active LOW)

Keypad (4x4) → Port 1

- Rows → P1.24, P1.25, P1.26, P1.27
- Columns → P1.28, P1.29, P1.30, P1.31

ADC (Pulse Sensor)

- Sensor Output → P0.28 (AD0.1)

THEORY AND WORKING PRINCIPLE:

This system is a real-time health monitoring system using LPC2148 microcontroller.

1. Pulse Measurement (BPM)

- Pulse sensor gives analog signal
- LPC2148 ADC converts it to digital value
- BPM calculated using:

$$BPM = \frac{ADC_value \times 120}{1023}$$

- Displayed on LCD as:
 - LOW (<60)
 - NORMAL (60–100)
 - HIGH (>100)

2. Temperature Input (Keypad)

- User enters temperature manually using keypad
- 2-digit input (like 36, 37, etc.)
- Converted using atoi()
- Displayed as:
 - LOW (<36°C)
 - NORMAL (36–37°C)
 - HIGH (>37°C)

3. LCD Display

- First line → BPM
- Second line → Temperature
- LEDs ON (blinking) if:
 - BPM > 100 OR
 - Temperature > 37°C
- Otherwise LEDs OFF

4. LED Alert System

- LEDs ON (blinking) if:
 - BPM > 100 OR
 - Temperature > 37°C
- Otherwise LEDs OFF

5. Keypad Working

- Column scanning method used
- Detects pressed key and returns value

TOOL & TECHNOLOGY USED:

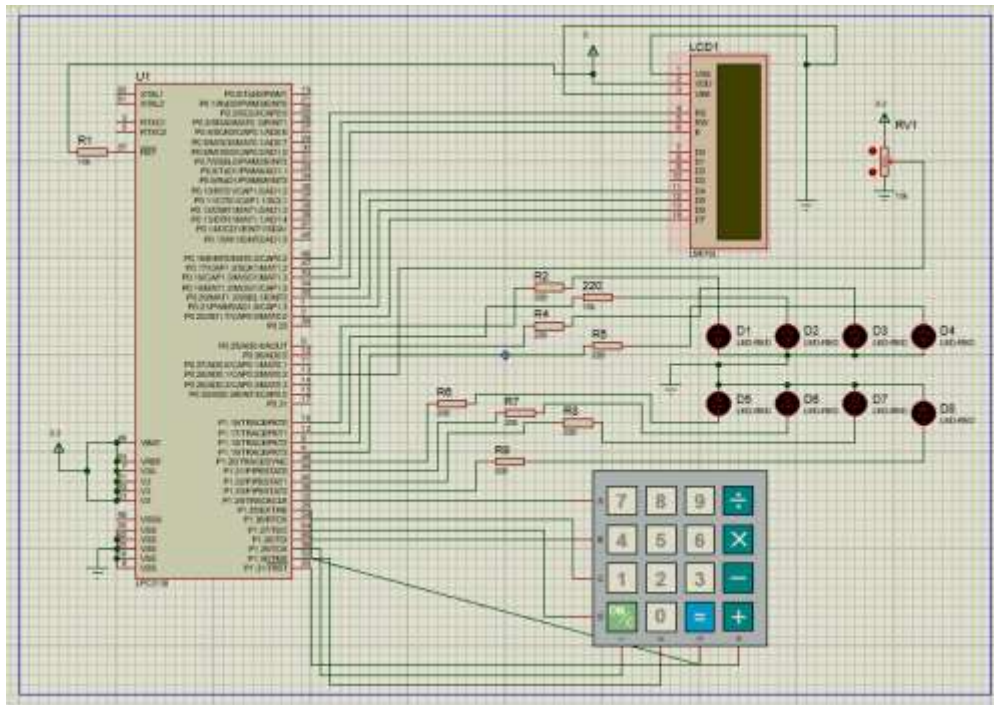
Hardware:

- LPC2148 Development Kit
- 6x2 LCD
- 4x4 Keypad
- LEDs

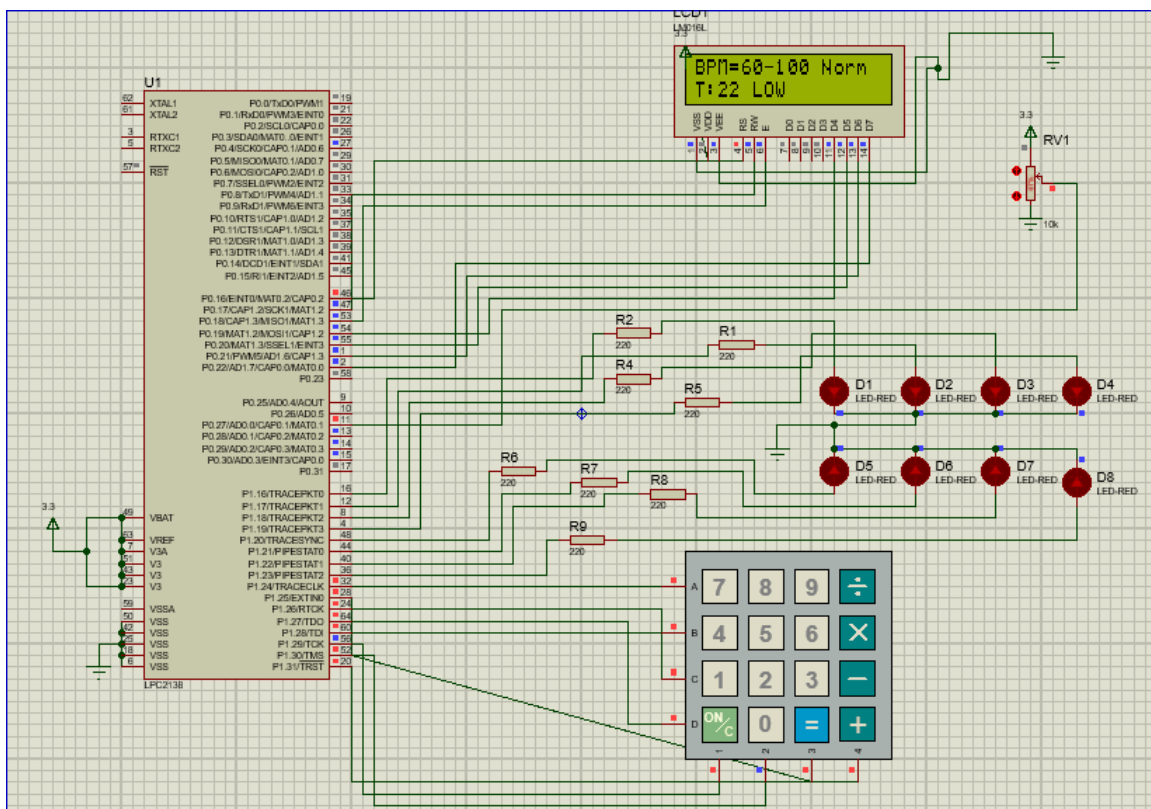
Software:

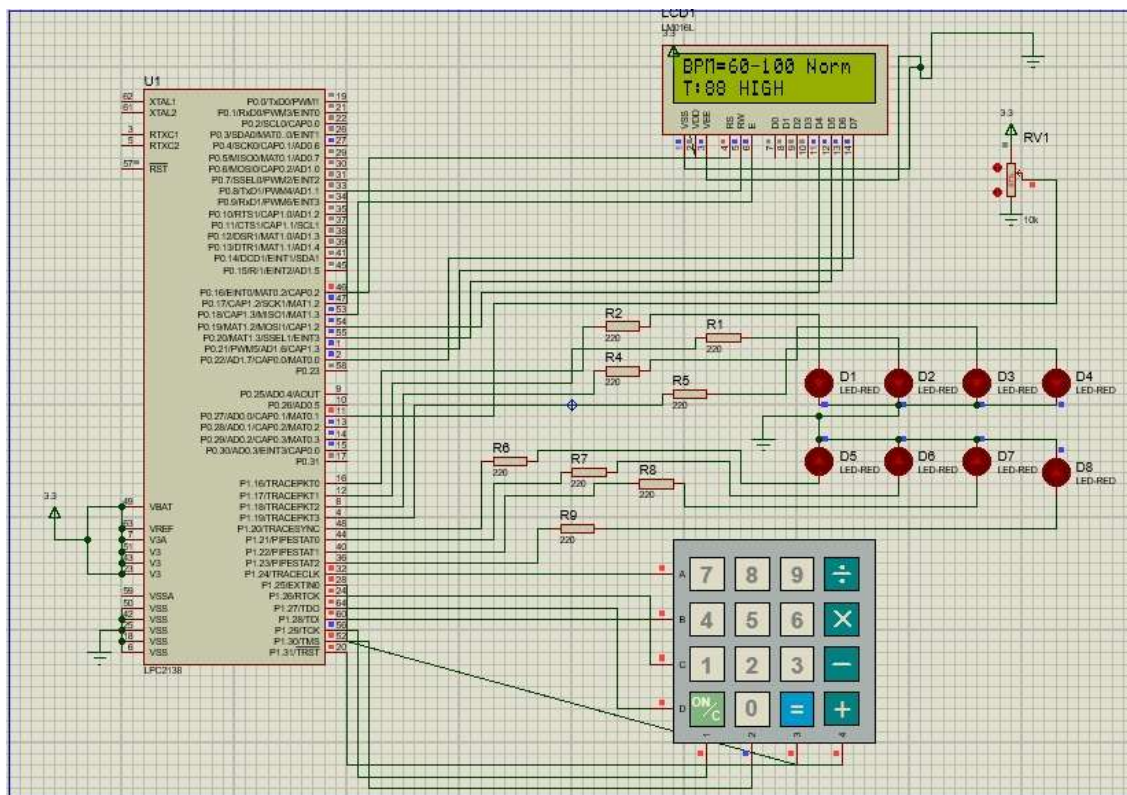
- Keil microVision
- Proteus (for simulation)

SCREENSHOT OF THE SIMULATOR (PROTEUS) BASED CIRCUIT IMPLEMENTATION:



SCREENSHOT OF THE OUTPUT WAVEFORM OBTAINED FROM THE SIMULATOR:





SCREENSHOT OF THE CIRCUIT BASED ON HARDWARE KIT:



SCREENSHOT OF THE OUTPUT OBTAINED FROM THE HARDWARE FOR THE CIRCUIT:

Pulse low:



Pulse normal:



Pulse high:



Temperature low:



Temperature normal:



Temperature high:



SINGLE VIDEO LINK FOR THE WORKING MODEL OF THE SOFTWARE CIRCUIT AND HARDWARE CIRCUIT (MUST BE A MICROSOFT LINK FROM THE FOLDER SHARED BY YOUR RESPECTIVE FACULTY):

RELEVANT INFORMATION:

1. System works in real-time
2. Uses ADC for analog sensing
3. Uses keypad for manual input
4. LCD provides user interface
5. LED gives emergency alert

CONCLUSION:

The project successfully demonstrates a basic health monitoring system using LPC2148. It measures pulse rate using ADC and accepts temperature input via keypad. The system provides real-time display and alert mechanism using LEDs.

PROBLEMS FACED DURING THE IMPLEMENTATION:

1. ADC value fluctuation
2. Keypad debouncing issues
3. LCD timing issues
4. Incorrect port configuration

PROBLEMS RESOLVED USING SOLUTIONS:

1. Added delay for stable ADC reading
2. Used proper keypad scanning logic
3. Corrected LCD initialization sequence
4. Verified pin connections

REFERENCES:

1. LPC2148 Datasheet
2. Keil Documentation
3. Microcontroller Lab Manual